

ECE 445 - AI Use

Jayden Aoki, Alexander Chinen

For this project we primarily used AI to efficiently process our datasets and generate our code. We used Anthropic's Claude AI and the Alu AI given with the course. We also used these models to better understand and apply the fundamental machine learning concepts that we learned in class. Below are the AI prompts that provided us with the most success.

AI Prompts

“Me and my partner want to make a machine learning project that takes data from our campus grid in totalized meters and tries to forecast future grid load depending on projected weather conditions. How can we start and what type of concepts/math will we need to use”

“create a model that will forecast a daily load profile graph”

“ give us a plan based on python and explain the math involved assuming I know nothing about ML”

“Given the totalized meter data of the college campus given in 15 min intervals, we want to create code for grid forecasting. Where can I get weather data for the UH manoa campus 2024-2025”

Details About the code

The final code is provided in the **“meter_forecast_project(2).zip”** file. The file should have the 5-steps to the machine learning algorithm, as well as the 15min interval meter data called **“substations_kw_15_min_fy25.csv”**